

PB

PolyBack

User Training Guide

Quantitative Research & Backtesting Platform
for Prediction Market Exchanges

Polymarket • Kalshi • Manifold

Version 1.0 | March 2026

Quick-Start Checklist

Follow these steps to get PolyBack running and execute your first backtest:

1. Start the backend server: `uvicorn backend.main:app --reload --port 8000`
2. Start the frontend: `cd frontend && npm run dev`
3. Open your browser at `http://localhost:5173`
4. Select your target exchange: Polymarket, Kalshi, or Manifold
5. Browse markets on the Backtest screen — search and filter by category
6. Click a market card to load its full probability history
7. Add 1–5 markets to the backtest queue using the + button
8. Select a strategy from the dropdown in the right-side backtest panel
9. Adjust the parameter sliders to tune entry/exit thresholds
10. Click Run Backtest and review equity curve, total return, Sharpe, and drawdown
11. Save the run to the History panel via the clock icon in the header
12. Iterate: change strategy or parameters and run again



No API Keys Needed to Start

All three exchanges expose read-only market data publicly — no credentials required to browse markets or run backtests. API keys are only needed if you want to place live orders.

1. Technical Requirements

1.1 System Requirements

Component	Minimum	Recommended
OS	Linux, macOS, Windows 10+ (WSL2)	Linux / macOS
RAM	4 GB	8 GB
Disk Space	500 MB free	2 GB
Network	Required (live API calls)	Low-latency connection

1.2 Backend Dependencies

Dependency	Version	Notes
Python	3.11+	3.12 recommended
FastAPI	0.110+	Auto-installed via pip
Uvicorn	0.29+	ASGI server
httpx	0.27+	Async HTTP client
pydantic	2.x	Data validation

Install backend dependencies and start the server:

```
cd ~/quant_project/Polymarket
pip install -r backend/requirements.txt
uvicorn backend.main:app --reload --port 8000
```

API Documentation

Once the backend is running, full interactive API docs are available at <http://localhost:8000/docs> — useful for debugging and integration testing.

1.3 Frontend Dependencies

Dependency	Version	Notes
Node.js	18+	20 LTS recommended
npm / pnpm	9+ / 8+	Either package manager works
Vite	5.x	Dev server and bundler
React	18.x	UI framework
TypeScript	5.x	Type checking

Install and start the frontend:

```
cd frontend
npm install
npm run dev
```

The application will be available at <http://localhost:5173>

1.4 Environment Configuration

Create a `.env` file in the `backend/` directory. You can also manage credentials through the Settings panel in the UI (recommended).

```
# Polymarket (optional – public endpoints require no key for reading)
POLY_API_KEY=your_polymarket_api_key
POLY_API_SECRET=your_polymarket_api_secret
POLY_PASSPHRASE=your_polymarket_passphrase

# Kalshi (optional – public endpoints are unauthenticated)
KALSHI_API_KEY=your_kalshi_api_key
KALSHI_API_PASSWORD=your_kalshi_api_password
```

Exchange	Account Required	Trading API
Polymarket	No (read) / Yes (trade)	CLOB — requires API key + secret + passphrase
Kalshi	No (read) / Yes (trade)	REST v2 — requires API key + password
Manifold	No (read or trade)	REST v0 — fully public, play-money (Mana)

✓ Safe for Prototyping

Manifold uses play money (Mana) and requires no authentication for any endpoint — making it the ideal exchange to prototype strategies without financial risk.

2. Glossary of Quantitative Terms

2.1 Core Concepts

Prediction Market A financial exchange where participants trade contracts that resolve to \$1 (YES) or \$0 (NO) based on real-world outcomes. The current price equals the crowd's implied probability.

Probability (p) In prediction markets, price and probability are equivalent. A YES contract at 0.35 means a 35% chance of resolving YES. All PolyBack prices are expressed as probabilities between 0 and 1.

Binary Contract A contract with exactly two outcomes: YES (\$1) or NO (\$0). PolyBack filters for binary markets exclusively for their well-defined payoff structures.

AMM (Automated Market Maker) A liquidity mechanism (used by Polymarket and Manifold) where a smart contract always quotes both buy and sell prices. Price adjusts automatically based on trade volume.

CLOB (Central Limit Order Book) Traditional market structure (used by Kalshi) where limit orders are matched by price-time priority. Provides explicit bid/ask prices and queue depth.

2.2 Price & Market Structure

Term	Definition
Bid	Highest price a buyer will pay for a YES share — immediate fill price for sellers
Ask	Lowest price a seller will accept — immediate fill price for buyers
Spread	Gap between bid and ask (ask - bid). Represents the round-trip cost of entering and exiting
Midpoint	$(\text{bid} + \text{ask}) / 2$ — best fair-value estimate when no last-trade price is available
Liquidity	Total capital deployed in a market. Higher = tighter spreads, less slippage
Slippage	Difference between expected and actual fill price due to market impact from large orders
Volume	Total notional traded. High-volume markets have better price discovery and tighter spreads

2.3 Strategy & Signal Concepts

Term	Definition
Entry Threshold	Price level at which a strategy opens a position (e.g., buy when $p < 0.15$)
Exit Threshold	Price level at which an open position is closed for profit (e.g., close when $p > 0.60$)
Stop Loss	Hard price floor — position is forcibly closed to cap losses (e.g., exit if p falls to 0.05)

Term	Definition
Edge	Difference between your true probability estimate and the market implied probability. Source of profit.
Alpha	Returns above a passive strategy. Generated by superior probability estimation or structural advantages.
Mean Reversion	Tendency of prices to return toward their average after a deviation (common after news-driven spikes)
Momentum	Tendency of prices to continue in the same direction — common as markets approach resolution

2.4 Statistical Concepts

Term	Definition
Z-Score	$z = (p_{\text{current}} - \mu) / \sigma$. How many standard deviations the current price is from the rolling mean
Rolling Window	Fixed-length lookback that slides forward with each data point. Controls recency vs. history sensitivity.
Standard Deviation (sigma)	Dispersion measure of a price series. Used to normalize entry thresholds in Z-Score strategies.
Sharpe Ratio	$(\text{mean_return} - \text{risk_free}) / \sigma$. Above 1.0 = good, above 2.0 = excellent.
Maximum Drawdown	Largest peak-to-trough decline during the backtest. Critical for assessing tail risk.
Win Rate	% of trades that closed profitably. Must be combined with reward/risk ratio to assess profitability.
Total Return	$(\text{final_equity} - \text{initial_capital}) / \text{initial_capital} * 100$. Net gain/loss over the backtest period.

2.5 Position Sizing

Kelly Criterion Mathematically optimal fraction of bankroll to maximize long-run growth: $f^* = (bp - q) / b$. Requires a reliable estimate of true probability.

Half-Kelly Betting $f^*/2$ instead of f^* . Reduces variance by ~75% with only ~25% reduction in growth rate. Strongly recommended in practice due to estimation error.

Recommendation

Always start with Fractional Kelly ($k = 0.25$ to 0.5) rather than Full Kelly. Overestimating your edge with Full Kelly can cause catastrophic drawdowns.

2.6 Execution Concepts

Mode	Description
Confirm	Each signal requires manual approval before any order is sent
Auto	Signals execute automatically without any manual intervention
Paper	Signals are logged and P&L tracked, but no real orders are placed (simulation)

3. Embedded Strategies Reference

PolyBack ships with seven built-in strategies. All parameters are configurable via the Strategies screen without writing code.

3.1 Threshold

Attribute	Value
Category	Mean Reversion
Risk Level	Low
Complexity	Simple

The simplest strategy: buy when probability falls below a fixed entry level, sell when it recovers above a fixed exit level. A hard stop-loss guards against tail events.

Signal Logic

```
Entry: p < T_entry AND no open position
Exit:  p > T_exit OR p < T_stop_loss
Size:  max(min_capital, available_liquidity / 10)
```

Parameter	Default	Description
Entry Threshold	0.15	Buy when probability drops below this level
Exit Threshold	0.60	Sell when probability rises above this level
Stop Loss	0.05	Hard exit if probability falls below this

✓ Best Use

Exploits short-lived dislocations in large, liquid markets where probabilities revert after news-driven shocks. Best for markets with a clear fundamental anchor and >30 days to resolution.

⚠ Key Risk

Fixed thresholds do not adapt to changing market dynamics. A genuine negative update can push price permanently below the entry threshold. Requires manual calibration per market category.

3.2 Momentum Chaser

Attribute	Value
Category	Trend Following
Risk Level	Medium

Attribute	Value
Complexity	Moderate

Captures upward probability trends early and exits on the first signs of reversal. Effective in markets where information cascades gradually — political outcomes, economic data releases, or breaking news.

Signal Logic

Entry: $(p_t - p_{\{t-1\}}) / p_{\{t-1\}} > \text{epsilon}$ AND $p_t < T_{\text{entry}}$
 Exit: price drops > trailing_stop% from peak OR $p_t < \text{stop_loss}$
 Size: Kelly fraction of estimated edge

Parameter	Default	Description
Entry Threshold	0.40	Minimum probability to enter a momentum position
Trailing Stop %	0.60	Exit when price drops by this % from peak
Stop Loss	0.05	Hard floor stop-loss

✓ Best Use

Early positioning into trending markets captures flow from late participants. Most effective in the final 2 weeks before market resolution as probability accelerates toward certainty.

⚠ Key Risk

False breakouts generate whipsaw losses. High sensitivity to noise in low-volume markets. Performance degrades in range-bound, mean-reverting regimes.

3.3 Z-Score Reversion

Attribute	Value
Category	Statistical Arbitrage
Risk Level	Medium
Complexity	Advanced

Maintains a rolling window of recent probabilities and computes a z-score to identify statistically abnormal dislocations. Enters when probability is significantly depressed relative to recent history.

Signal Logic

$z = (p_t - \mu_{\text{window}}) / \sigma_{\text{window}}$
 Entry: $z < -z_{\text{entry}}$ AND no open position

```
Exit:  z >= z_exit   OR   z < -z_stop (runaway – stop out)
Size:  |z_t| * base_position * volatility_adjustment
```

Parameter	Default	Description
Window Size	20	Number of ticks in rolling window
Entry Z-Score	1.5	Enter when z-score drops below negative this value
Exit Z-Score	0.0	Exit when z-score reverts to this level
Stop Z-Score	3.0	Hard stop if z-score falls below negative this value

✓ Best Use

More adaptive than fixed-threshold strategies — entry auto-adjusts as market volatility changes. Most effective in markets with >30 days to resolution.

⚠ Key Risk

Z-score assumes stationarity. Markets approaching resolution violate this assumption as probabilities accelerate toward 0 or 1. Short windows are noisy; long windows miss short-term opportunities.

3.4 Kelly Criterion

Attribute	Value
Category	Position Sizing
Risk Level	High
Complexity	Advanced

Uses the Kelly formula to size each trade proportional to estimated edge. Bets more when the edge is large and less when uncertain. Requires a reliable estimate of your true probability.

Formula

```
f* = (bp - q) / b
b = (1 - p_true) / p_true
stake = k * f* * bankroll   (k = kelly_fraction, e.g. 0.5 for Half-Kelly)
```

Parameter	Default	Description
Kelly Fraction	0.5	Fraction of full Kelly (0.5 = Half-Kelly, recommended)
Entry Threshold	0.20	Market price must be below this to enter
Exit Threshold	0.65	Probability target to close position
Stop Loss	0.05	Hard floor stop-loss

⚠ Key Risk

Edge estimate error is catastrophic — overestimating p_{true} leads to severe overbetting. Full Kelly ($k=1.0$) produces extreme volatility. Always use Half-Kelly ($k=0.5$) or lower.

3.5 Mean Reversion

Attribute	Value
Category	Statistical Arbitrage
Risk Level	Medium
Complexity	Moderate

Fades extreme probability moves by taking opposing positions when price deviates more than k standard deviations from the rolling mean. Uses absolute deviation rather than a normalized z-score.

Parameter	Default	Description
Lookback Window	15	Rolling window for mean/std calculation
Reversion Threshold	2.0	Standard deviations from mean to trigger entry

3.6 Market Making

Attribute	Value
Category	Liquidity Provision
Risk Level	Low-Medium
Complexity	Advanced

Posts bids below short-term fair value and exits when price rises by the full spread. Generates consistent small wins from spread collection rather than directional prediction.

Parameter	Default	Description
Spread Width	0.04	Target bid-ask spread to collect (4 cents)
Stop Loss	0.08	Hard loss floor per position

3.7 Structure Harvest (Coming Soon)

Attribute	Value
Category	Event-Driven
Risk Level	High
Complexity	Expert

Targets systematic wealth transfers from takers to makers documented across tens of millions of prediction market trades. Takers consistently overpay due to affirmative bias; this strategy posts limit orders on the maker side waiting for emotional taker flow.

 Development Status

Structure Harvest is under active development. Parameters and full logic will be published upon release.

3.8 Creating Custom Strategies

The Strategies screen includes a + Custom Strategy button. Custom strategies can be:

- Named, described, and tagged with a category and risk level
- Assigned any number of parameters with labels, defaults, min/max/step
- Edited later via the Edit button in the strategy detail header
- Deleted if no longer needed

 Tip

Custom strategies participate in all backtest and signal-generation workflows alongside built-in strategies. Create variants of built-in strategies with different defaults to quickly compare approaches.

4. Screen Walkthroughs

The top header contains a tab bar with six views. The active view is highlighted in teal. The exchange selector (Polymarket / Kalshi / Manifold) applies globally — switching exchanges reloads all market data.

4.1 Backtest Screen

The primary workspace for strategy research. The layout has three columns: market search on the left, market detail in the center, and the backtest configuration panel on the right.

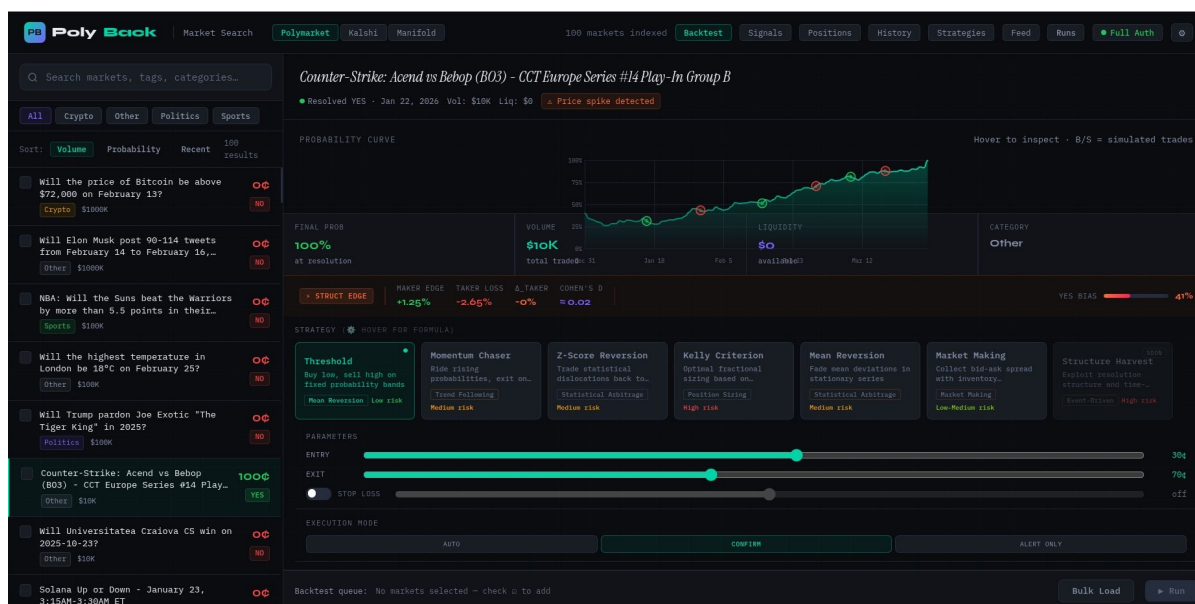


Figure 1 — Backtest screen: market list (left), probability curve detail (center), strategy selector and parameter sliders (right)

Left Column — Market Search

- Search box: Filter markets by title text in real time
- Category filter: Narrow by Politics, Economics, Crypto, Sports, etc.
- Sort control: Order by volume, liquidity, probability, or recency
- Market cards: Each card shows title, current probability (color-coded), volume, and end date

Center/Right — Market Detail

Once a market is selected, the detail panel shows:

- Title, exchange badge, resolution status, volume, and liquidity
- Probability curve: interactive chart of full price history — hover for exact price at any point
- Edge metrics: Maker Edge, Taker Loss, Cohen's D, and YES Bias
- Strategy cards: all available strategies shown with entry conditions and risk levels
- Parameter sliders: live-adjustable for the selected strategy
- Execution mode toggle: Confirm, Auto, or Alert Only

Running a Backtest

13. Click the + button on market cards to add them to the backtest queue
14. Use Bulk Add to quickly add the top-N markets by volume
15. Select a strategy from the dropdown
16. Adjust parameter sliders — changes apply to the next run
17. Click Run Backtest — results appear below with equity curve, trade log, and per-market stats

✓ Batch Testing

Queue up to 20+ markets and run a single backtest to see aggregate and per-market performance in one pass. Use the clock icon to save the run to your History.

4.2 Signals Screen

Displays live trade signals generated by active strategies, along with a real-time execution log.

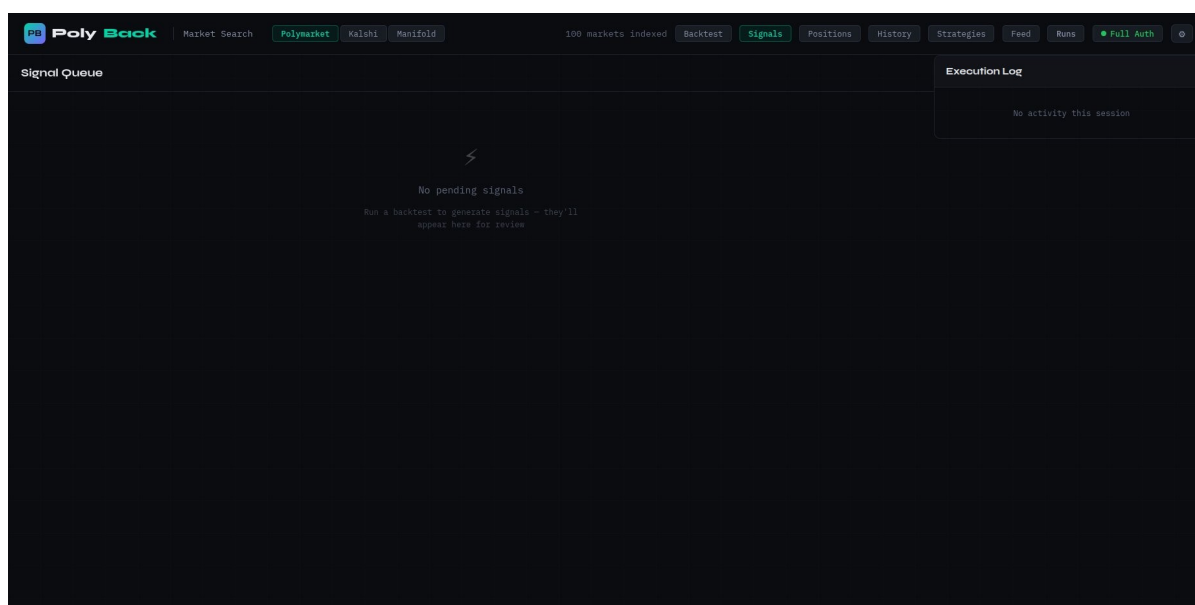


Figure 2 — Signals screen showing the Signal Queue (left) and Execution Log (right)

Signal Queue

Each signal card shows: market title and exchange, direction (YES/NO), entry price, suggested size, generating strategy with rationale, confidence score (0-100%), and Confirm/Dismiss action buttons.

⚠ Execution Mode

The execution mode toggle at the top controls whether signals require manual approval (Confirm) or execute automatically (Auto). Always use Confirm mode until you have validated a strategy's live performance.

Execution Log

A scrolling log of all executions and dismissals with timestamps, prices, and outcomes. Useful for auditing strategy behavior over time.

4.3 Positions Screen

Tracks all currently open positions across all strategies and markets in real time.

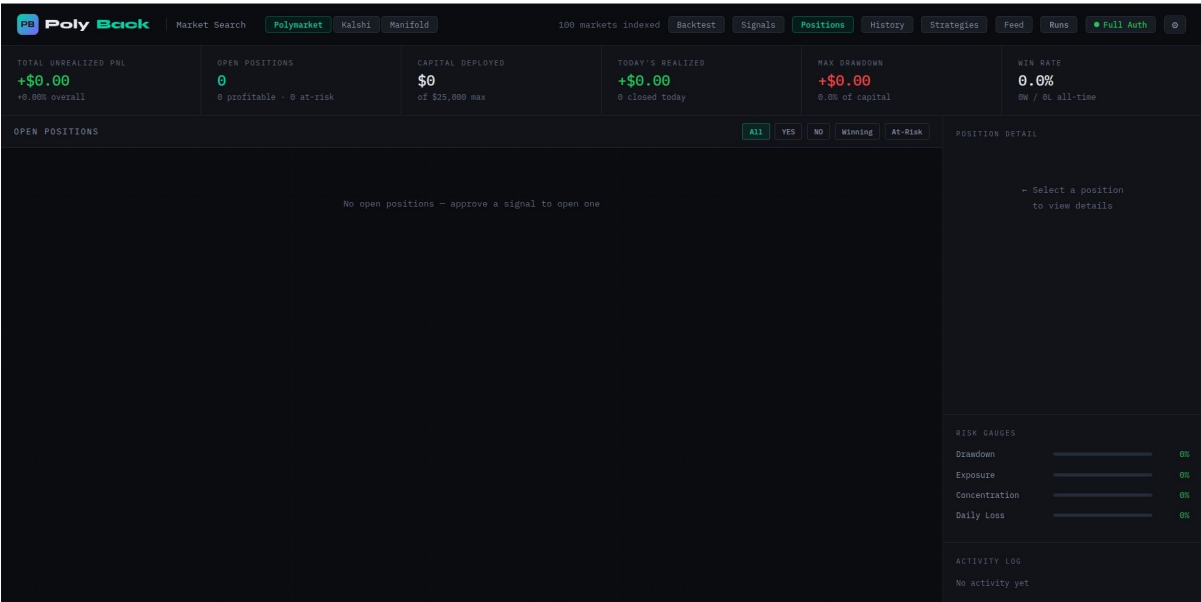


Figure 3 — Positions screen with risk gauges (Drawdown, Exposure, Concentration, Daily Loss) in the right panel

Each position row shows: market and exchange, direction (YES/NO) and entry price, current live-updating price, unrealized P&L in dollars and %, generating strategy, and time since entry.

Metric	Description
Total Unrealized PnL	Aggregate floating gain/loss across all open positions
Open Positions	Count of positions — profitable vs. at-risk breakdown
Capital Deployed	Total capital currently at work, vs. maximum configured
Max Drawdown	Largest unrealized decline from peak capital
Win Rate	Running win/loss ratio across all positions (all-time)

i Risk Gauges

The right panel shows four live risk gauges: Drawdown, Exposure, Concentration, and Daily Loss. Monitor these continuously during live trading — they provide an at-a-glance view of portfolio stress.

4.4 History Screen

A searchable archive of all completed backtest runs from the current session.



Figure 4 — History screen showing cumulative PnL curve, strategy PnL breakdown, and filterable trade log

Each run entry shows: run date/time, strategy used, markets tested, and aggregate results (average return, win rate, Sharpe). Clicking a run expands it to show full per-market results and equity curves.

- Cumulative PnL area chart: portfolio value over the session
- PnL by Strategy bar chart: compare performance across strategies
- Strategy breakdown table: per-strategy trades, PnL, and win rate
- Trade log: every individual entry and exit, filterable by strategy and exit reason

✓ History Drawer

The clock icon in the top-right header opens a compact list view for quickly switching between runs without leaving the current view.

4.5 Strategies Screen

The strategy library and configuration center. All available strategies are displayed as cards.

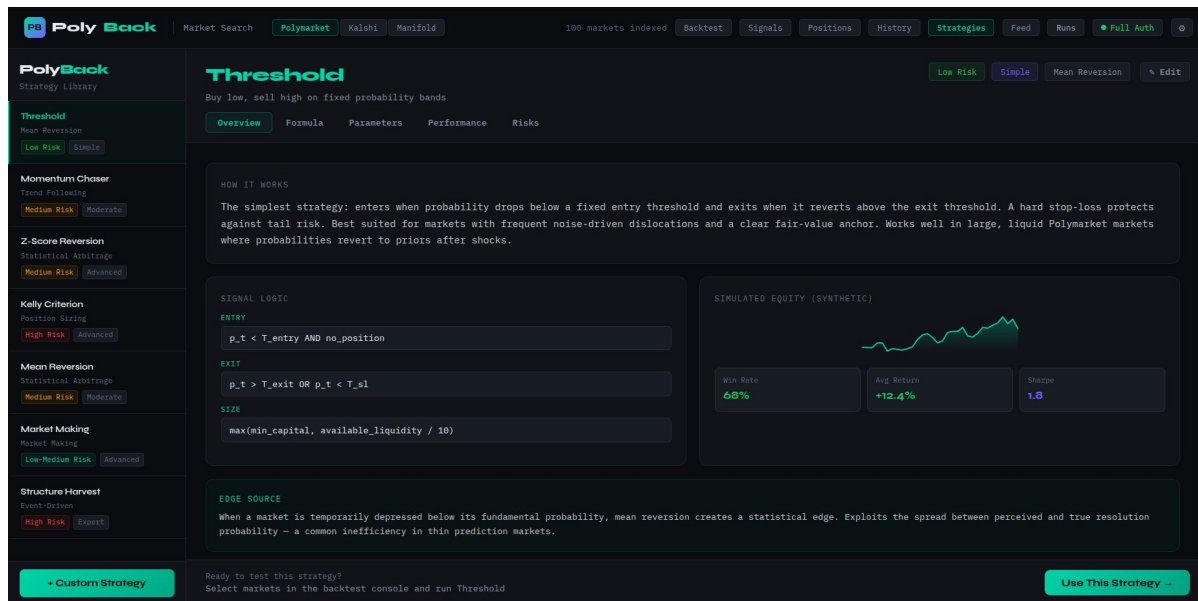


Figure 5 — Strategies screen showing the library (left sidebar) and Strategy Detail Panel with signal logic, formula, and synthetic equity curve

Each strategy card shows: name and tagline, category/risk level/complexity badges, synthetic equity curve preview, and key performance stats.

Strategy Detail Panel

Clicking a strategy card opens the detail view with: full description and edge rationale, entry/exit logic pseudocode, mathematical formula, parameter definitions, known risks, and historical performance metrics.

Editing and Creating Strategies

18. Click Edit (pencil icon) in the detail header to modify any strategy field
19. Click + Custom Strategy (top-right) to create a new strategy from scratch
20. Fill in name, description, parameters, category, and risk level
21. Click Save Strategy — it immediately appears in the library and backtest selector
22. Delete custom strategies via the trash icon (built-in strategies cannot be deleted)

4.6 Feed Screen

A real-time market data feed showing live prices, recent trades, and order book snapshots. Each market card shows current probability with trend direction, recent price change, bid/ask spread, and volume ticker. The feed auto-refreshes every 10 seconds, sorted by most recently traded first.

Order Book

Selecting a market opens a live order book view. For CLOB exchanges (Kalshi) this shows real depth; for AMM exchanges (Polymarket, Manifold) it shows a synthetic spread based on pool depth.

4.7 Settings Panel

Accessible via the gear icon in the top-right header. Click the backdrop or X to close.

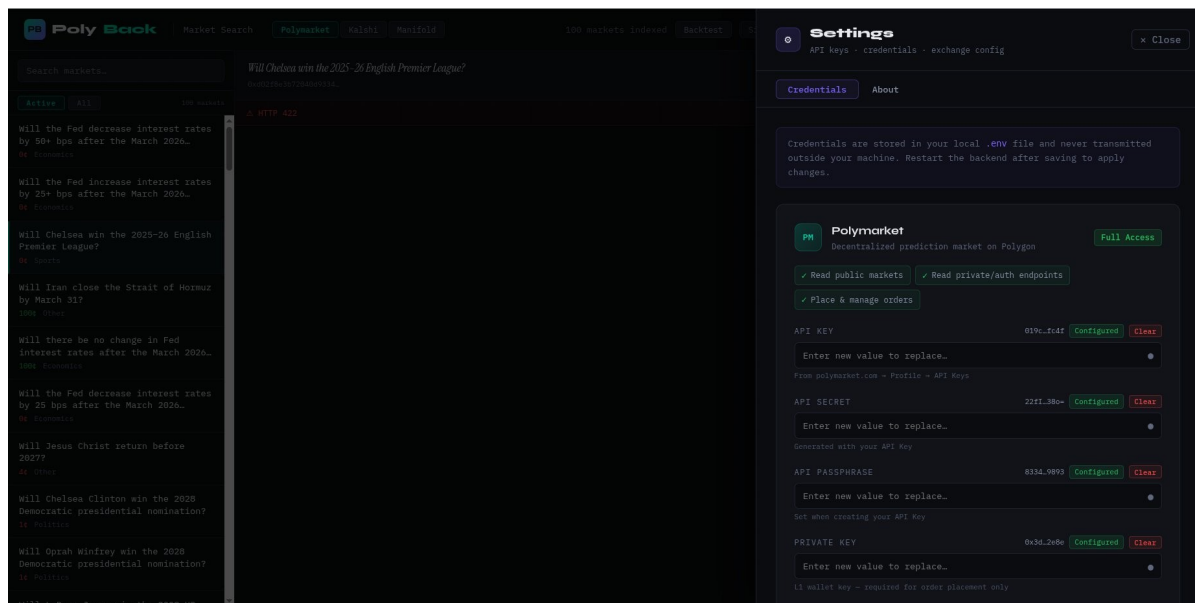


Figure 6 — Settings panel showing credential configuration for Polymarket with masked key previews and status badges

Credentials Tab

Manage API keys for each exchange. Each exchange section shows:

- Auth badge: green 'Configured' if keys are set, gray 'Not Set' if not
- Capabilities grid: what the exchange supports (read markets, price history, order book, trading)
- Credential fields: one row per required key field with show/hide toggle

23. Type your key in the password field (click the eye icon to show/hide)

24. Click Save to write the key to the .env file on the backend

25. Click Clear to remove a stored key

26. A masked preview (first 4 + last 4 characters) confirms the key is set

✓ Security

API keys are stored server-side in .env and are never returned to the browser. The UI only receives a configured: true/false flag and the masked preview. Your keys never leave your machine.

DISCLAIMER

PolyBack is a research and analysis tool. Nothing in this guide constitutes financial advice. Prediction market trading involves risk of loss. Always use Paper mode and Fractional Kelly sizing until you have thoroughly validated a strategy's live performance.